
Fourier Analysis on Finite Groups

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January, 2023

Abstract

This essay briefly introduces the applications of representations and characters to harmonic analysis, while blending the analytic and algebraic viewpoints. We shall first give a quick review of some useful materials learned in class, and give some generalizations of them. Then we focus on Fourier analysis on finite abelian groups, and give a systematic treatment of Dual groups, Pontryagin duality and Fourier transform. We also introduce a striking application of finite Fourier analysis to number theory. Next we discuss the non-abelian case, and prove Wedderburn's theorem. We culminate the paper with a brief and naive glimpse of abstract harmonic analysis, and see some general results on locally compact abelian groups.

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1 Introduction

We first introduce some notations and list some useful results learned in class and their corollaries.

1.1 The structure of $\mathbb{C}G$

We use $\mathbb{C}G$ to denote the group algebra of a finite group G , but we shall interpret it as the set of all complex valued functions on G , which is a more “analytic” point of view. The identity of $\mathbb{C}G$ is the map δ_1 that maps 1 to 1 and vanishes elsewhere. The product in $\mathbb{C}G$, with our notation, gives rise to the definition of convolution:

Definition (Convolution)

For $f_1, f_2 \in \mathbb{C}G$, the convolution of f_1, f_2 is defined by

$$(f_1 * f_2)(g) = \sum_{h \in G} f_1(gh^{-1})f_2(h)$$

It’s easy to see the convolution is simply a restatement of our original product in $\mathbb{C}G$. Take $G = \mathbb{Z}_n$ for example, $\mathbb{C}\mathbb{Z}_n$ is exactly $\mathbb{Z}_n[x]$, there are two ways to calculate the product of polynomials, one can proceed term by term, or group all terms of the same degree together. It also illuminating that the discrete convolution takes analogous form with the classic convolution on Euclidean spaces, with sum taking place of integration.

$\mathbb{C}G$ is also endowed with an inner product given by

$$\langle f, g \rangle := \sum_{g \in G} f_1(g)\overline{f_2(g)}$$

The norm it induces makes $\mathbb{C}G$ into a Banach algebra.

1.2 Unitary Representations

We say a representation ρ on G is unitary if $\rho(g)$ is unitary for all $g \in G$. The important facts is the following:

Proposition

Every representation of a finite group G is equivalent to a unitary representation

Proof. We again use the crucial “averaging trick”: let $\langle -, - \rangle$ be the standard inner product on $\mathbb{C}^n$, we define a new inner product:

$$(u, v) = \sum_{g \in G} \langle \rho(g)u, \rho(g)v \rangle$$

It’s straightforward to check that $(-, -)$ is sesquilinear and positive definite. Now we find

that

$$(\rho(g)u, \rho(g)v) = (u, v)$$

This says $\rho(g)$ is unitary in our new inner product. Let $(u, v) = u^* P^t P v$ with real orthogonal P , then we have $\rho(g)^* P^t P \rho(g) = I$, thus $\rho(g)$ is similar to some unitary matrix thru P . $\square$

1.3 Characters as Basis

Let $C_j (j = 1, \dots, r)$ be the conjugacy classes of G , and use $Z(\mathbb{C}G)$ to denote the set of class functions (functions that is constant on conjugacy classes). In class we have proved that, characteristic function of each conjugacy class $\overline{C_j}$ constitute a basis of $\mathbb{C}G$, that is to say $Z(\mathbb{C}G)$ is also the center of $\mathbb{C}G$.

Let $\chi_j (j = 1, \dots, r)$ be the distinct irreducible characters of G , A corollary of schur's lemma is the orthogonality of irreducible characters, with the row orthogonality saying they form an orthogonal set in $\mathbb{C}G$:

$$\langle \chi_i, \chi_j \rangle = |G| \delta_i^j$$

We know that

$$\dim Z(\mathbb{C}G) = r = \dim \text{Span} \{ \chi_j \} \leq \dim Z(\mathbb{C}G)$$

hence $\{ \chi_j \}$ is also an orthogonal basis of $Z(\mathbb{C}G)$. For $f \in Z(\mathbb{C}G)$,

$$f = \frac{1}{|G|} \sum_j \langle f, \chi_j \rangle \chi_j$$

If G is abelian, $Z(\mathbb{C}G) = \mathbb{C}G$, whence $\{ \chi_j \}$ form a basis of $\mathbb{C}G$.

One might wonder what is the basis of $\mathbb{C}G$ in general. Let

$$\rho^k = (\rho_{i,j}^k), G \rightarrow GL_{d_k}(\mathbb{C}), k = 1, \dots, r$$

be non-equivalent irreducible and unitary representations on G .

Peter-Weyl Theorem

The

$$\sum_{k=1}^r d_k^2 = |G|$$

matrix indices

$$\left\{ \frac{d_k}{|G|} \rho_{i,j}^k \mid i, j = 1, \dots, d_k, k = 1, \dots, r \right\}$$

constitute an orthonormal basis of $\mathbb{C}G$

Proof. Let $\rho \sim V$ and $\eta \sim W$, let the standard basis: $e_i, e_j \in V$ and $f_k, f_l \in W$, consider the linear transformation

$$Tv = \sum_{g \in G} \eta(g^{-1}) \langle \rho(g)v, e_j \rangle f_l$$

then $T\rho(g) = \eta(g)T$. If two representations are non-equivalent, so $T = 0$ by Schur's lemma, i.e.

$$\begin{aligned} \langle Te_i, f_k \rangle &= \sum_{g \in G} \langle \rho(g)e_i, e_j \rangle \langle f_l, \eta(g) f_k \rangle \\ &= \sum_{g \in G} \rho_{ij} \overline{\eta_{kl}} \\ &= 0 \end{aligned}$$

On the contrary, if $\rho = \eta$, $\deg \rho = d$, $e_k = f_k, e_l = f_l$, by Schur's lemma $T = \alpha I$, but

$$\frac{d\alpha}{|G|} = \text{tr } T = \text{tr } (\langle -, e_j \rangle e_l) = \delta_{jl}$$

and

$$\frac{|G|}{d} \delta_{ik} \delta_{jl} = \langle Te_i, e_k \rangle = \sum_{g \in G} \rho_{ij} \overline{\eta_{kl}}$$

As a consequence, $\left\{ \frac{d_k}{|G|} \rho_{ij}^k \right\}$ is an orthonormal set. Note that

$$\dim \mathbb{C}G = \dim \text{Span} \left\{ \frac{d_k}{|G|} \rho_{ij}^k \right\} = |G|$$

we're done. □

2 Fourier Analysis on Finite Abelian Groups

We now turn our attention to finite Fourier analysis, where infinite sums and integrals in classical analysis are replaced by finite sums, thus the theory is particularly elegant and simple with questions of measure and convergence and disappear.

2.1 The Dual Group and Pontryagin Duality

For an abelian group G , the group algebra $\mathbb{C}G$ is also commutative. All irreducible characters are linear characters. In fact, they are homomorphisms into the unit circle $\mathbb{S}$ (as subgroup of $\mathbb{C}^*$), since the finite subgroup of $\mathbb{C}^*$ is also a subgroup of $\mathbb{S}$. More specifically, G can be written as direct product $C_{n_1} \times \cdots \times C_{n_r}$, let g_k be the generator of C_{n_k} , the irreducible characters take the form of

$$g_1^{i_1} \cdots g_{i_r}^{i_r} \mapsto w_{i_1}^{i_1} \cdots w_{i_r}^{i_r}$$

where w_k is n_k th root of unity.

We have learned in class that the product of two linear character is again an irreducible character. This suggests a natural group structure:

Definition (Dual Group)

The set $\text{Hom}(G, \mathbb{S})$ of all irreducible characters on G , denoted as $\widehat{G}$, is called the dual group of G . Define the operation in $\widehat{G}$ by pointwise product $(\chi \cdot \eta)(g) := \chi(g)\eta(g)$

For example, let $G = \mathbb{Z}_n$, then

$$\widehat{G} = \left\{ \chi_k : [m] \mapsto e^{\frac{2\pi i k m}{n}} \mid k = 0, \dots, n-1 \right\}$$

which is isomorphic to G by $\chi_k \mapsto [k]$. We shall show this is true in general.

Proposition

G is isomorphic to its bidual:

$$G \cong \widehat{\widehat{G}}$$

we call G a self-dual group.

Proof. If G_1, G_2 are abelian groups, $\text{Hom}(-, \mathbb{S}) : \text{Ab} \rightarrow \text{Ab}$ is a (contravariant) representation functor, hence preserves the direct product

$$\text{Hom}(G_1 \times G_2, \mathbb{S}) \cong \text{Hom}(G_1, \mathbb{S}) \times \text{Hom}(G_2, \mathbb{S})$$

thus

$$\widehat{G_1 \times G_2} \cong \widehat{G_1} \times \widehat{G_2}$$

For any abelian group G , it splits into a direct product of cyclic groups: $G \cong C_{n_1} \times \dots \times C_{n_r}$, thus $\widehat{G} \cong \widehat{C_{n_1}} \times \dots \times \widehat{C_{n_r}}$. Since $C_n \cong \widehat{C_n}$, we obtain:

$$G \cong \widehat{\widehat{G}}$$

□

However, this isomorphism depends on the choice of the generators of each cyclic group, hence not coordinate free. A more intrinsic correspondence is the isomorphism between G and its bidual $\widehat{\widehat{G}}$.

Theorem (Pontryagin Duality)

The map given by

$$\varphi : G \rightarrow \widehat{\widehat{G}}, g \mapsto (\varphi_g : \chi \mapsto \chi(g))$$

is an isomorphism of groups.

Proof. For $\chi, \eta \in \widehat{G}$, $\varphi_g(\chi\eta) = \chi(g)\eta(g) = \varphi_g(\chi)\varphi_g(\eta)$, thus the map is well defined. Besides, for $g, h \in G$, $\varphi_{gh}(\chi) = \chi(g)\chi(h) = \varphi_g\varphi_h$, thus a homomorphism. The map is injective since $\varphi_g(\chi) = 1$ for all $\chi \in \widehat{G}$ implies

$$[G : C_g] = |C_G(g)| = \sum_{\chi \in \widehat{G}} \chi(g)\overline{\chi(g)} = |G|$$

thus $g = 1$. It follows that φ is an isomorphism. $\square$

Clearly the bidual functor $\varphi = \text{Hom}(\text{Hom}(-, \mathbb{C}), \mathbb{C})$ is covariant, with φf defined by: $\varphi f : \varphi_g \mapsto \varphi_{f(g)}$. Pontryagin theorem states that the natural transformation α between the identity functor and the bidual functor is an isomorphism.

$$\begin{array}{ccc} G & \xrightarrow[\cong]{\alpha_G: g \mapsto \varphi_g} & \varphi G \\ \downarrow f & & \downarrow \varphi f \\ H & \xrightarrow[\cong]{\alpha_H: h \mapsto \varphi_h} & \varphi H \end{array}$$

Pontryagin duality allows us to identify G with $\widehat{\widehat{G}}$, thus we may simply write $g(\chi) = \chi(g)$.

2.2 Fourier Transform

In analysis, we discuss Fourier transform with two lines in mind, one for the original function and another for its transform. In finite Fourier analysis, we have two groups in mind, one is G , and another is its dual group $\widehat{G}$. We shall use $\widehat{G}$ as the underlying set of the Fourier transform.

Definition (Fourier Transform)

The Fourier transform of $f : G \rightarrow \mathbb{C}$, denoted as $\widehat{f} : \widehat{G} \rightarrow \mathbb{C}$, is defined by

$$\widehat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g)\overline{\chi(g)}$$

In the case of cyclic groups, $\widehat{G}$ is trivially isomorphic to G , which allows us to define the Fourier transform for cyclic group from $\mathbb{C}G$ to itself. For $f : \mathbb{Z}_n \rightarrow \mathbb{C}$, define $\widehat{f} : \mathbb{Z}_n \rightarrow \mathbb{C}$ by

$$\widehat{f}([m]) = n \langle f, \chi_m \rangle = \sum_{k=0}^{n-1} f([k])e^{-2\pi i m k / n}$$

For any abelian group A , there are two operations make $\mathbb{C}A$ into an algebra: convolution and pointwise multiplication. The identity for convolution is the delta function δ_1 , while for pointwise

multiplication is the map with constant value 1. At first glance, Fourier transform links two Banach algebras $(\mathbb{C}G, +, *)$ and $(\mathbb{C}\widehat{G}, +, \cdot)$ together by $\mathcal{F} : \mathbb{C}G \rightarrow \mathbb{C}\widehat{G}, f \mapsto \widehat{f}$. We shall see that $\mathcal{F}$ is in fact an isomorphism. First we show that it's injective:

Theorem (Fourier Inversion)

If $f \in \mathbb{C}G$, then

$$f = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi$$

Proof. This is just a restatement of the fact that $\widehat{G}$ form a basis of $\mathbb{C}G = Z(\mathbb{C}G)$:

$$f = \sum_{\chi} \langle f, \chi \rangle \chi$$

□

We define an inner product $(-, -)$ in $\mathbb{C}\widehat{G}$ by

$$(g, h) = \frac{1}{|G|} \sum_{\chi} g(\chi) \overline{h(\chi)}$$

for $g, h \in \mathbb{C}\widehat{G}$ (The coefficient $1/|G|$ is exactly “dual measure” to the counting measure.) Thus, the column orthogonality

$$(g_i, g_j) = \frac{|C_G(g_i)|}{|G|^2} \delta_i^j = \frac{1}{|C_i||G|} \delta_i^j$$

for $g_i, g_j \in \widehat{G}$ implies that $G = \widehat{\widehat{G}}$ form an orthogonal basis of $\mathbb{C}\widehat{G}$. The Fourier Inversion Theorem is then

$$f(g) = (\langle f, - \rangle, g^{-1})$$

for $f \in \mathbb{C}G$ and $g \in G$. If one simply define the Fourier transform of $h \in \mathbb{C}\widehat{G}$ by

$$\widehat{h} = (h, -)$$

we can write

$$\widehat{\widehat{f}}(g) = f(g^{-1})$$

A direct corollary of Fourier Inversion Theorem is that $\mathcal{F}$ preserves the inner product:

$$\langle f_1, f_2 \rangle = (\widehat{f_1}, \widehat{f_2})$$

Particularly, we have the Parseval-Plancherel formula:

$$\|f\|_G^2 = \frac{1}{|G|^2} \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 = \|\widehat{f}\|_{\widehat{G}}^2$$

Clearly $\mathcal{F}$ is surjective, this is because $\dim \mathbb{C}G = \dim \mathbb{C}\widehat{G} = |G|$. Besides, $c_1 \widehat{f_1 + c_2 f_2} = c_1 \widehat{f_1} + c_2 \widehat{f_2}$. It remains to show that $\mathcal{F}$ preserves multiplication:

Convolution Theorem

For $a, b \in \mathbb{C}G$:

$$\widehat{a * b} = \widehat{a} \cdot \widehat{b}$$

Proof.

$$\begin{aligned} \widehat{(a * b)}(\chi) &= \sum_{g, h \in G} a(gh^{-1})b(h)\overline{\chi(g)} \\ &= \sum_{h, \sigma \in G} a(\sigma)b(h)\overline{\chi(\sigma h)} \\ &= \sum_{h, \sigma \in G} a(\sigma)\overline{\chi(\sigma)}b(h)\overline{\chi(h)} \\ &= \widehat{a}(\chi) \cdot \widehat{b}(\chi) \end{aligned}$$

□

Thus, Fourier transform gives an isomorphism between two Banach algebras $(\mathbb{C}G, +, *, \langle -, - \rangle)$ and $(\widehat{\mathbb{C}G}, +, \cdot, \langle -, - \rangle)$.

2.3 An Application to Number Theory

We shall make a detour to introduce a striking application of Fourier analysis of finite groups to the Dirichlet's theorem on primes: the number of the primes of the form $kq + l, k \in \mathbb{Z}$ is infinite if $(k, l) = 1$. we outline the main steps of the proof, and omit the tedious details.

One prove that $\{p \equiv l\}$ is infinite by showing that $\sum_{p \equiv l} \frac{1}{p^s}$ deiverges for $s = 1$. In this particular case, it's the theory of Fourier series on $\mathbb{Z}_q^*$ that plays a key role in the solution of the problem.

Let φ be the Euler phi-function, then $|\mathbb{Z}_q^*| = \varphi(q)$. Consider the characteristic function of l on $\mathbb{Z}_q^*$:

$$\delta(l) = \begin{cases} 1 & n \equiv l \pmod{q} \\ 0 & \text{otherwise} \end{cases}$$

Let $G = \mathbb{Z}_q^*$. Expand δ_l in a Fourier series

$$\delta_l(n) = \sum_{e \in \widehat{G}} \widehat{\delta}_l(e) e(n)$$

where

$$\widehat{\delta}_l(e) = \frac{1}{|G|} \sum_{g \in G} \delta_l(g) \overline{e(g)} = \frac{1}{|G|} \overline{e(l)}$$

We extend the function δ_l to $\mathbb{Z}$ by setting $\delta_l(m) = 0$ if $(m, q) \neq 1$. Besides, we shall extend $e \in \widehat{G}$

to $\mathbb{Z}$ by

$$\chi(m) = \begin{cases} e(m) & (m, q) = 1 \\ 0 & \text{otherwise} \end{cases}$$

χ is called the dirichlet charachters modulo q . The trivial character is χ_0 as the characteristic function of $\{m : (m, q) = 1\}$. It's easy to see χ is also a multiplicative. Now the expansion reads

$$\delta_l(m) = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(l)} \chi(m)$$

Thus

$$\begin{aligned} \sum_{p \equiv l} \frac{1}{p^s} &= \sum_p \frac{\delta_l(p)}{p^s} \\ &= \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(l)} \sum_p \frac{\chi(p)}{p^s} \\ &= \frac{1}{\varphi(q)} \sum_p \frac{\chi_0(p)}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(l)} \sum_p \frac{\chi(p)}{p^s} \\ &= \frac{1}{\varphi(q)} \sum_{p \nmid q} \frac{\chi_0(p)}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(l)} \sum_p \frac{1}{p^s} \end{aligned}$$

We know by analysis on Riemann zeta function that $\sum_p \frac{1}{p}$ diverges, thus the first item diverges as $s \rightarrow 1$. It remains to show the second item is bounded. Thus, Dirichlet Theorem is reduced to the boundedness of

$$\sum_p \frac{\chi(p)}{p^s}$$

as $s \rightarrow 1$. The multiplicativity of χ implies that

$$L(s, \chi) \sum_n \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \chi(p)/p^s}$$

Now take logarithm of the product, we have

$$\log L(s, \chi) = \sum_p \frac{\chi(p)}{p^s} + O(1)$$

Now we reduce the theorem to the non-vanishing of L-functions at $s = 1$.

3 The Non-abelian Case

For non-abelian group G , $\mathbb{C}G$ with convolution is also non-commutative, and G will no-longer isomorphic with $\widehat{G}$, thus one must find another appropriate underlying space for the Fourier transformation. The characters no longer form a group, and they do not span $\mathbb{C}G$, but the intrices of irreducible unitary representations: $\{\rho_{ij}^k\}$ form an orthogonormal basis for $\mathbb{C}G$. Thus one can define Fourier transform for non-abelian case by:

$$\widehat{f}(\rho_{ij}^k) = \sum_{g \in G} f(g) \overline{\rho_{ij}^k(g)}$$

or in the matrix form

$$\widehat{f}(\rho^k) = \sum_{g \in G} f(g) \overline{\rho^k(g)}$$

The Peter-Weyl theorem implies that

$$f = \frac{1}{|G|} \sum_{i,j,k} d_k \widehat{f}(\rho_{ij}^k) \rho_{ij}^k$$

i.e. the Fourier inversion formula stands valid. It's easy to verify the linearity of the transform.

More over, the convolution theorem remains true:

$$\begin{aligned} \widehat{(a * b)}(\rho) &= \sum_{g,h \in G} a(gh^{-1}) b(h) \overline{\rho(g)} \\ &= \sum_{h,\sigma \in G} a(\sigma) b(h) \overline{\rho(\sigma h)} \\ &= \sum_{h,\sigma \in G} a(\sigma) \overline{\rho(\sigma)} b(h) \overline{\rho(h)} \\ &= \widehat{a}(\rho) \cdot \widehat{b}(\rho) \end{aligned}$$

Combining the above facts, we have

Theorem (Wedderburn)

The Fourier transform is an isomorphism between the algebras

$$\begin{aligned} \mathcal{F} : \mathbb{C}G &\rightarrow GL_{d_1}(\mathbb{C}) \times \cdots \times GL_{d_r}(\mathbb{C}) \\ f &\mapsto (f(\rho^1), \dots, f(\rho^r)) \end{aligned}$$

If we define on the algebra $GL_{d_1}(\mathbb{C}) \times \cdots \times GL_{d_r}(\mathbb{C})$ an inner product

$$(a, b) := \frac{1}{|G|^2} \sum_{k=1}^r d_k \operatorname{tr}(a(\rho^k) b(\rho^k)^*)$$

then one can easily prove that

$$\langle a, b \rangle = (\widehat{a}, \widehat{b})$$

Thus, $\mathcal{F}$ still gives an isomorphism between Banach algebras.

4 A Glimpse of Abstract Harmonic Analysis

We briefly discuss some basic results in Abstract Harmonic Analysis, and see how to recover the discrete case from the general theory (though we merely scratch the surface of this rich theory.)

Let $\mathbf{LCA}$ be the category of locally compact group, the morphisms are continuous homomorphisms. The dual group of $G \in \mathbf{LCA}$ is defined as $\widehat{G} := \operatorname{Hom}_{\mathbf{LCA}}(G, \mathbb{S})$, or equivalently, the set of unitary characters of G , as a version of Schur's lemma still holds for locally compact group. $\widehat{G}$ is an abelian group under pointwise multiplication. Besides, if we give $\widehat{G}$ the compact-open topology, then $\widehat{G} \in \mathbf{LCA}$. Generally, a group G is not isomorphic to the dual group $\widehat{G}$. For example, $\widehat{\mathbb{Z}} = \mathbb{S}$, $\widehat{\mathbb{S}} = \mathbb{Z}$. However, the Pontryagin theorem remains valid in $\mathbf{LCA}$.

Theorem (Pontryagin)

There is a canonical (functorial) isomorphism $G \cong \widehat{\widehat{G}}$ for any $G \in \mathbf{LCA}$ given by

$$g \mapsto (\chi \mapsto \chi(g))$$

There are many proofs of the theorem in the literature. A fancy proof of Roeder [4] uses only category approachs and homological method without involving measure theory and Fourier analysis. The only deep result used is Peter-Weyl theorem in $\mathbf{LCA}$.

The Pontryagin duality is strongly analogous to the cananical isomorphism between finite dimensional linear space or reflexive Banach space. In fact, in an extensive class of infinite-dimensional topological linear spaces the Pontrjagin theorem is valid, and in reflexive linear topological spaces the group-duality theorem is also true and, in fact, merely restates the reflexive property [5].

To conduct Fourier analysis on G , we have to replace the sum by integration, thus an appropriate measure m on the Borel σ -algebra $\mathcal{B}$ of G is introduced. The measure should be invariant under translation:

$$m(S) = m(gS)$$

Besides, m should be locally bounded, and satisfies the regularity condition

$$m(S) = \inf \{m(U) : S \subset U, U \text{ open}\}$$

$$m(V) = \sup \{m(K) : K \subset V, K \text{ compact}\}$$

for $S \in \mathcal{B}$ and V open. The measure satisfies these conditions is called Haar measure. Haar's Theorem asserts that every group $G \in \mathbf{LCA}$ pocesses a Haar measure, and it's unique up to a positive constant. The integrable functions $L^1(G)$ is an commutative algebra with multiplication defined by convolution:

$$(f * g)(\sigma) := \int_{h \in G} f(\sigma h^{-1})g(h)$$

Definition (Fourier Transform)

For $f \in L^1(G)$, the Fourier transform is

$$\widehat{f}(\chi) = \int_G f \overline{\chi}$$

For $g \in G$, Pontryagin duality gives a unique element $\varphi_g : \chi \mapsto \chi g$ in $\widehat{G}$, we may identifying G with $\widehat{\widehat{G}}$ and omit writing φ ., i.e. $g(\chi) := \chi(g)$. The results in classical Fourier analysis remains valid:

Theorem

1. Fourier transform gives a homomorphism from $L^1(G)$ into $C^0(\widehat{G})$, it's linear and Convolution Theorem holds:

$$\widehat{f * g} = \widehat{f} \cdot \widehat{g}$$

2. Fourier inversion formula holds in $L^1(G)$:

$$f(g) = \int_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(g) \text{ a.e.}$$

for a unique Haar measure $\widehat{m}$ on $\widehat{G}$.

3. The uniqueness Theorem: $\widehat{f}_1 = \widehat{f}_2$ implies $f_1 = f_2$.

4. Besides, Fourier transform extends to an isomorphism of $L^2(G)$ and $L^2(\widehat{G})$, with Plancherel Theorem holds :

$$\int_G |f|^2 = \int_{\widehat{G}} |\widehat{f}|^2$$

where the measure on $\widehat{G}$ is $\widehat{m}$.

We call the unique Haar measure on $\widehat{G}$ such that the Inversion Formula holds the dual measure on $\widehat{G}$. The inversion formula can also be written as

$$\widehat{\widehat{f}}(g) = f(g^{-1})$$

A corollary of Pontryagin duality is that, $\widehat{G}$ is compact iff G is discrete, where if the Haar measure on G is chosen s.t. $m(G) = 1$, the dual measure is the counting measure. Conversely, $\widehat{G}$ is discrete iff G compact, and G taken the counting measure then the dual measure is the one s.t. $\widehat{m}(\widehat{G}) = 1$.

For instance, $\widehat{\mathbb{Z}} = \mathbb{S}$, we recover the theory of Fourier series: the measure on $\mathbb{Z}$ is the counting measure and the measure on $\mathbb{S}$ is the Lebesgue measure $d\theta/2\pi$. If $G = \mathbb{R}^n$, $\widehat{G} = \mathbb{R}^n$, both choose the Lebesgue measure $1/2\pi^{n/2}dx$. If G is finite, we choose the counting measure on G , while the measure on $\widehat{G}$ must be $1/|G| \times$ counting measure.

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